

Conjugate prior

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In [Bayesian inference](#), the prior distribution of a parameter and the likelihood of the observed data are combined to obtain the posterior distribution of the parameter.

If the prior and the posterior belong to the same parametric family, then the prior is said to be conjugate for the likelihood.



Review of parametric families

First of all, let us review the concept of a [parametric family](#).

Let a set of probability distributions be put in correspondence with a [parameter space](#).

If the correspondence is a function (i.e., it associates one and only one distribution in to each [parameter](#)), then is called a parametric family.

Examples of parametric families

Examples of parametric families are:

- the set of all [normal distributions](#), indexed by the parameter vector , where and are the mean and the variance of the distribution;
- the set of all [exponential distributions](#), indexed by the rate parameter ;
- the set of all [chi-square distributions](#), indexed by the degrees-of-freedom parameter .

Prior, likelihood and posterior

In a Bayesian inference problem, we specify two distributions:

- the likelihood, that is, the distribution of the observed data conditional on the parameter :
- the [prior distribution](#) of the parameter

After observing the data, we use [Bayes' rule](#) to compute the [posterior distribution](#)

$$p(\theta|x) = \frac{p(x|\theta)p(\theta)}{p(x)} = \frac{p(x|\theta)p(\theta)}{\int_{\theta} p(x|\theta)p(\theta)d\theta}$$

Definition of a conjugate prior

We can now define the concept of a conjugate prior.

Definition Let be a parametric family. A prior belonging to is said to be conjugate for the likelihood if and only if the posterior belongs to .

In other words, when we use a conjugate prior, the posterior resulting from the Bayesian updating process is in the same parametric family as the prior.

Example

In the lecture on [Bayesian inference about the mean of a normal distribution](#), we have already encountered a conjugate prior.

In that lecture, $\mathbf{x}$ is a vector of [IID draws](#) from a normal distribution having unknown mean μ and known [variance](#) σ^2 . Moreover, both the prior and the posterior distribution of the parameter μ are normal. Hence, the prior and the posterior belong to the same parametric family of normal distributions, and the prior is conjugate with the likelihood.

Usefulness of conjugate priors

There are basically two reasons why models with conjugate priors are popular (e.g., [Robert 2007](#), [Bernardo and Smith 2009](#)):

1. they usually allow us to derive a closed-form expression for the posterior distribution;
2. they are easy to interpret, as we can easily see how the parameters of the prior change after the Bayesian update.

Exponential families

Conjugate priors are easily characterized when the distribution of $\mathbf{x}$ belongs to an [exponential family](#), in which case the likelihood takes the form

where:

- the base measure $\mu(\cdot)$ is a function that depends only on the $\mathbf{x}$ vector of data ;
- the parameter θ is a η vector;
- the sufficient statistic $T(\mathbf{x})$ is a vector-valued function of $\mathbf{x}$;
- the log-partition function $A(\theta)$ is a function of θ .
- $\eta^T T(\mathbf{x})$ is the dot product between η and $T(\mathbf{x})$.

A parametric family of conjugate priors for the above likelihood is formed by all the distributions such that

where:

- η is a η vector of parameters;
- ν is a scalar parameter;
- $k(\chi, \nu)$ is a function that returns the normalization constant needed to make $p(\theta|\mathbf{x})$ a proper probability density (or mass) function.

The parameters η and ν are called hyperparameters.

Note that

implies that

As a consequence, the above parametric family of conjugate priors, called a **natural family**, contains all the distributions associated to couples of hyperparameters (η, ν) such that the integral in the denominator is well-defined and finite.

Given the likelihood and the prior, the posterior is

$$p(\theta|\mathbf{x}) = k(\chi + T(\mathbf{x}), \nu + 1) \exp[\theta^T (\chi + T(\mathbf{x})) - (\nu + 1)A(\theta)]$$

provided $k(\chi + T(\mathbf{x}), \nu + 1)$ is well-defined.

Proof

Examples of natural families

Let us now see how powerful the technology of natural families is, by deriving the conjugate priors of some common distributions.

Bernoulli likelihood and beta priors

Remember that a [Bernoulli random variable](#) is equal to 1 with probability q and to 0 with probability $1-q$.

Suppose that we observe a realization x of the Bernoulli variable and we want to carry out some Bayesian inference on the unknown parameter q .

The likelihood has exponential form:

$$\begin{aligned} p(x|q) &= 1_{\{x \in \{0,1\}\}} q^x (1-q)^{1-x} \\ &= 1_{\{x \in \{0,1\}\}} \left(\frac{q}{1-q} \right)^x (1-q) \\ &= 1_{\{x \in \{0,1\}\}} \exp \left[x \ln \left(\frac{q}{1-q} \right) + \ln(1-q) \right] \\ &= 1_{\{x \in \{0,1\}\}} \exp \left[x \ln \left(\frac{q}{1-q} \right) - \ln \left(1 + \exp \left(\ln \left(\frac{q}{1-q} \right) \right) \right) \right] \\ &= h(x) \exp[\theta^T T(x) - A(\theta)] \\ &= p(x|\theta) \end{aligned}$$

where $1_{\{x \in \{0,1\}\}}$ is an indicator function equal to 1 if $x \in \{0,1\}$ and to 0 otherwise, and

The natural family of conjugate priors contains priors of the form

Since $\eta(q)$ is an increasing function of q and

$$\begin{aligned} \frac{d\theta}{dq} &= \frac{d}{dq} \ln \left(\frac{q}{1-q} \right) \\ &= \frac{1-q}{q} \left[\frac{1}{1-q} + \frac{q}{(1-q)^2} \right] \\ &= \frac{1}{q} + \frac{1}{1-q} \\ &= \frac{1-q+q}{q(1-q)} = \frac{1}{q(1-q)} \end{aligned}$$

we can apply the [formula for the density of an increasing function](#):

$$\begin{aligned} p(q) &\propto \exp \left[\ln \left(\frac{q}{1-q} \right) \chi + \nu \ln(1-q) \right] \frac{1}{q(1-q)} \\ &= \left(\frac{q}{1-q} \right)^{\chi} (1-q)^{\nu} \frac{1}{q(1-q)} \\ &= q^{\chi-1} (1-q)^{\nu-\chi-1} \end{aligned}$$

Thus, the natural family of conjugate priors contains priors that assign to q a [Beta distribution](#) with parameters χ and ν .

According to the general formula derived above for natural families, the posterior distribution of q is

$$\begin{aligned} p(\theta|x) &\propto \exp[\theta^T (\chi + T(x)) - (\nu + 1)A(\theta)] \\ &= \exp[\theta(\chi + x) - (\nu + 1)A(\theta)] \end{aligned}$$

which implies (by the same argument just used for the prior) that the posterior distribution of q is

that is, a Beta distribution with parameters $\chi + x$ and $\nu + 1$.

Poisson likelihood and Gamma prior

If X has a [Poisson distribution](#), its likelihood is

where λ is a parameter and x is the set of non-negative integer numbers.

We can write the likelihood in exponential form:

$$\begin{aligned} p(x|\lambda) &= 1_{\{x \in \mathbb{Z}_+\}} \exp(-\lambda) \frac{\lambda^x}{x!} \\ &= 1_{\{x \in \mathbb{Z}_+\}} \frac{1}{x!} \exp[\ln(\lambda)x - \exp(\ln(\lambda))] \\ &= h(x) \exp[\theta^T T(x) - A(\theta)] = p(x|\theta) \end{aligned}$$

where

The natural family of conjugate priors contains priors of the form

Since $\ln(\lambda)$ is an increasing function of λ and

we can apply the formula for the density of an increasing function:

Thus, the natural family of conjugate priors contains priors that assign to λ a [Gamma distribution](#) with parameters ν and χ .

By the general formula for natural families, the posterior distribution of λ is

$$\begin{aligned} p(\theta|x) &\propto \exp[\theta^T (\chi + T(X)) - (\nu + 1)A(\theta)] \\ &= \exp[\theta(\chi + x) - (\nu + 1)A(\theta)] \end{aligned}$$

which implies (by the same argument just used for the prior) that the posterior distribution of λ is

that is, a Gamma distribution with parameters $\nu + 1$ and $\chi + x$.

References

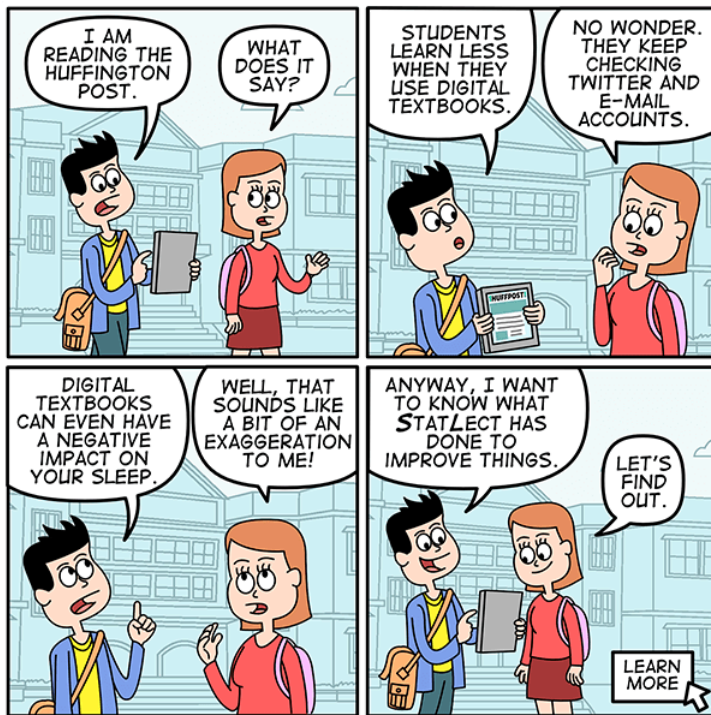
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